Project Name:

OceanJet Line

High-Speed Zero-Emission Public Transport and Infrastructure

The Report on the Public Consultation Research On The Possibility Of Implementing A Public Transport Ferry Service From Swansea To Destinations Across The Bristol Channel.

Response to UK Shared Prosperity Fund Research Grant CPR8/24/056

By David Sampson, CEO and team at Ocean Prime Industries Ltd

October 2026 (first issued March 2026)

Corrections and Clarifications

I regret that there were errors in the original report, which have now been corrected. I have updated some of the information where it is relevant. I have also removed some clumsy language that create some ambiguity in interpretation. (I would advise anyone using AI to analyse the report to note that “AI makes mistakes” and it doesn’t always interpret ambiguity in the best way).

It is important to take this report in context: It provides a level of evidence – the number of replies from the public – which is primarily in response to heavily congested roads. Thousands of people consider a ferry to be one possible solution. For the ferry design and propulsion technology, this should be considered as illustrative and pre-feasibility desk research, indicating that a ferry service using a hydrogen-electric hybrid is possible, but there are many considerable caveats, particularly as to when sufficient green hydrogen would be available.

Likewise, the economic benefits a ferry might bring to Swansea should be taken as an indicator of the potential scale, not a guaranteed outcome. Again, there are far too many unknowns to be precise. To be clear this is not a business case, nor should it be used as such.

This file (2026-10-26_OceanJet_Research_Report_Final) replaces the report issued in March 2026. The corrections and clarifications made in this version are set out below.

Section	March 2026 version	This version (October 2026)
Acknowledgement, Introduction, Executive Summary, 1.1	Response count given as 4,393, over 4,393, 4396 and over 4,396	4,396 wherever a specific figure is given. Approximate figures (more than 4,300, nearly 4,400) are unchanged
Executive Summary, 1.1	97.8% yes, 1.5% maybe, 0.7% no	97.7% yes, 1.5% maybe, 0.8% no, adjusted in line with the 4,396 total
Executive Summary	The research was commissioned by the City and County of Swansea Council and delivered between October 2024 and February 2025	The research was funded by a UK Shared Prosperity Fund research grant administered by Swansea Council. The research was developed between October 2024 and February 2025, and the public consultation opened in March 2025
Introduction	Millennium Pier project for Minehead, supported by the sci-fi author Isaac Asimov	Supporter corrected to the science fiction author Arthur C. Clarke, who was born in Minehead
Introduction	On a par with HS2	Comparable in strategic importance to HS2 between London and Birmingham
Vision	Arrival points at the harbour at Ilfracombe, Appledore or Burry Port	Ports named are indicative examples. All ports in the target zone are prospects, and no port selections or service decisions have been made. Ilfracombe is considered suitable for foot passengers, not vehicles

Section	March 2026 version	This version (October 2026)
1.1	No description of how the public learned of the consultation	Method added: the public learned of the research through a single press release, reported in the local press; every member of the public had the same opportunity to respond
1.3	In 2025 Storm Darragh produced waves over 9m	Storm date corrected to December 2024 (Note: Waves of over 9 meters were measured off Westward Ho! (at the Bideford Bay wave buoy) during Storm Darragh on December 7, 2024)
.2, Reference [5]	HGV signal-controlled system expected by October 2026	The HGV signal-controlled system on the M48 Severn Bridge has been delayed by National Highways and is now expected in January 2027
3.2, 3.3	Reduced service above 2.5m waves; target maximum wave height for comfortable operations 3+m	Clarified: 2.5m is the planning assumption for the schedule; 3+m is the vessel design target
3.3	We aim to build a small working prototype this year	The March 2026 report stated the intention to build a small working prototype in 2026. A small-scale prototype remains part of the vessel development programme, with timing dependent on funding and completion of the design
5.3	The model assumption is 50% of passengers arriving at Swansea would be new visitors and 50% returning residents.	The model assumption is 50% of all passengers would be visitors to Swansea and 50% returning residents.
5.3	Annual visitor economy benefit to Swansea could exceed £600 million	Clarified as estimates for the service as proposed at the time of the research: up to 800,000 visitors a year from a single vessel and up to £600 million a year from the full fleet. Actual numbers and spend depend on service details, passenger mix, disembarkation point, and local transport services and infrastructure, all yet to be agreed
6.1	Approximately 800,000 car journeys replaced, avoiding around 29,000 tonnes of CO2 per year	Car journeys corrected to 796,250 (1,592,500 passengers ÷ 2 per car). CO2 saving corrected to 28,900 tonnes per year (796,250 × 175 miles × 207.1 g/mile = 28,858 tonnes).
6.1	"…assuming every passenger would otherwise have driven"	Removed for ambiguity.

Section	March 2026 version	This version (October 2026)
Appendices	The following appendices contain supporting detail and are not intended for public release	Proprietary internal financial models, supporting spreadsheets, raw survey data and detailed assessments are confidential, held separately and excluded from public release. Any third-party disclosure requires Ocean Prime Industries Ltd's prior written approval and appropriate safeguards.

Acknowledgement

Ocean Prime Industries Ltd gratefully acknowledges the support of Swansea Council through the UK Government's UK Shared Prosperity Fund, which enabled the public consultation research detailed in this report. The grant (reference CPR8/24/056, value £24,999.95) funded the development of the research website and data collection infrastructure that generated 4,396 responses from across the Bristol Channel region.

Foreword by Rob Stewart

Leader of The City and County of Swansea Council

Swansea has always been a city with strong maritime connections. For generations, the Bristol Channel has been both a natural gateway and a barrier between communities on either side.

This report explores how modern technology and new thinking could help reconnect those communities in a cleaner, faster and more sustainable way.

Swansea Council supported this research through the UK Shared Prosperity Fund to better understand whether there is genuine public interest in a zero-emission passenger ferry linking Swansea with destinations across the Bristol Channel.

The response has been remarkable. More than 4,300 people and organisations took part in the consultation, with overwhelming support for exploring the idea further. What stands out most clearly from the findings is the level of enthusiasm for a transport option that could significantly reduce journey times by up to half or better while avoiding congestion on major road routes such as the M4 and M5.

People also recognise the importance of developing cleaner forms of transport that support our environmental ambitions and help tackle climate change.

For Swansea, projects like this represent more than just a transport connection. They offer the potential to strengthen economic ties with the South West of England, attract new visitors to our city, and support new industries linked to green maritime technology and hydrogen energy. Importantly, this report does not present a finished scheme. Instead, it provides an evidence base that will help inform the next stage of discussions with partners, industry experts and potential investors.

Delivering a project of this scale would require strong collaboration between the public and private sectors, as well as innovation in areas such as vessel design, port infrastructure and clean energy supply. I would like to thank Ocean Prime Industries Ltd for leading this research and everyone who contributed their views through the consultation.

Public engagement is essential when considering ambitious ideas like this, and the feedback received will play an important role in shaping what happens next.

Swansea has a proud history as a city that embraces innovation and looks outward to new opportunities. While there are many challenges to address, the findings in this report show there is strong public interest in exploring how a modern, zero-emission ferry service could play a role in the region's future transport network.

I look forward to seeing how this conversation develops in the months and years ahead.

Introduction by Dave Sampson

CEO Ocean Prime Industries Ltd

This project is no longer just about a ferry. In my opinion it is a nationally significant strategic infrastructure initiative, comparable in strategic importance to HS2 between London and Birmingham, that will provide high value jobs, economic growth and environmentally conscious reindustrialisation for coastal communities, where jobs mean a reason to stay and grow.

Thank you, Matt Callaghan, Greg Jones, Martin Nicholls, Rob Stewart, and everyone in the City and Council of Swansea County who have helped me research and complete this report. A special thanks to those in Council meetings who have for years kept the idea of a Swansea Ferry alive.

Thank you to members and their technical advisors and harbour masters of Torridge, North Devon, Neath Port Talbot and Somerset Councils for your energy and insights. They have contributed to my better understanding of the opportunities we might unlock and challenges we may encounter.

Thank you to the 4,396 people who have responded to the research so far. You have all helped shape the project. The research is still live on our website, and it is not too late to add your vote and comment, we continue to review them.

Since I first heard about the Swansea Ferry idea at a meeting on Floating Offshore Wind in North Devon College, in 2021, I have had conversations with hundreds of people on both sides of the Bristol Channel, most notably Chris Marrow the founder of the Severn Link, a strong supporter for this new initiative, whose enthusiasm and energy is extraordinary. I have learnt about the Millennium Pier project for Minehead, supported by the science fiction author Arthur C. Clarke, who was born in Minehead, and the extraordinary opportunities represented by the Blue Eden - Swansea Lagoon and the West Somerset Lagoon both offering decades of clean renewable tidal energy. I have met the team at the "Its Your Swansea" events in the Arena that are a powerful force and the many companies and organisations that they work with.

Thank you to everyone who reached out immediately following the research, providing marketing assistance, important historical material, suggested business collaboration, technical support, or offered support in other ways. I have had conversations about the possibility of expanding the project to other parts of the UK and abroad. Thank you again to each of you.

Finally thank you to the journalists and social media influencers who have promoted the story so far. The publicity unlocks opportunities.

Executive Summary

This report presents the findings from public consultation research into demand for a zero-emission ferry service connecting Swansea with destinations across the Bristol Channel and Celtic Sea. The research was funded by a UK Shared Prosperity Fund research grant administered by Swansea Council. The research was developed by Ocean Prime Industries Ltd between October 2024 and February 2025, and the public consultation opened in March 2025.

The response was exceptional. 4,396 individuals and organisations participated so far, with 97.7% (4296) expressing support for the proposed service, 1.5% (66) voted "maybe" and 0.8% (34) voted no. This represents one of the strongest public mandates for a transport infrastructure project in recent Welsh history.

Respondents consistently highlighted three themes: the desire for faster travel that avoids congested road routes, enthusiasm for zero-emission transport, and the potential economic benefits for communities on both sides of the channel. Concerns raised by those who were uncertain or opposed centred on commercial viability, weather and tidal challenges, and infrastructure requirements at destination ports.

Based on this research and market analysis, we believe there is substantial demand for a high-speed, affordable, zero-emission passenger service operating from Swansea and many other ports and harbours. We estimate target fares from £15 are possible and journey times of approximately one hour, the service would fundamentally change connectivity across the region.

Problem

Communities on opposite sides of the Bristol Channel can see each other's coastlines but cannot reach them. There is no scheduled passenger service connecting South Wales to Devon and Somerset. A journey from Swansea to Ilfracombe that should take an hour by sea takes four or five hours by road, most of it on congested motorways that are nowhere near either coast. For the communities at each end, this is not a minor inconvenience. It is a structural barrier to employment, tourism, healthcare, education, and social connection that has persisted for decades.

Vision

The research points to a different future. A zero-emission coastal passenger network designed to the standard people now expect from the best contemporary public transport: fast, frequent, affordable, fully accessible, and integrated into existing bus, rail, and active travel networks. The experience of London's Elizabeth line offers the closest comparison, not in scale, but in principle. The Elizabeth line succeeded because it was designed as a complete journey, not just a railway. Every element, from ticketing to wayfinding to the calm, light-filled stations themselves, was conceived as a single passenger experience, intuitive enough that first-time users need no instructions. It connects Heathrow to Paddington, Bond Street to Liverpool Street, making places that were always close but poorly linked suddenly feel part of the same city.

A coastal equivalent would do the same for the Bristol Channel: contactless ticketing, real-time information, level boarding, comfortable modern vessels, simple drive-on, drive-off boarding for cars. Pet and cyclist friendly, with crew trained to the best customer service standards, connecting waterfront arrival points that feel welcoming, well-designed, and appropriate to the places they serve, whether that is Swansea or indicative examples such as

Ilfracombe, Appledore or Burry Port. All ports in the target zone are prospects and no decisions on ports or services have been taken. Civic spaces, designed for the communities around them, are energy-positive where possible, following the principle already demonstrated at the Bay Technology Centre on Baglan Energy Park, the first commercial building in Wales that produces more energy than it consumes.

The benefits become real when you think about what the connections unlock. The Swansea Arena's 3,500-seat programme of concerts and events becomes accessible to audiences in Minehead, Bideford, Barnstaple, with late-evening return services after the show. The newly approved Skyline Swansea attraction on Kilvey Hill, Swansea's first international adventure tourism destination, and the Penderyn Copperworks Distillery on the Hafod Morfa site gain a catchment that extends well beyond the M4 corridor to the millions of summer visitors in South West England. Visitors arriving at Swansea can step onto the Copper Jack on the River Tawe, walk the Gower beaches, or use the newly proposed rail upgrades serving the region.

In the other direction, passengers from South Wales gain direct access to the Tarka Trail, the South West Coast Path, Dartmoor, and a part of England that is equally beautiful and equally underserved by public transport. And for people who simply need to get to work, to college, or to a hospital appointment on the other side of the water, the service becomes routine rather than impossible.

For people in North Devon, they would be able to take advantage of the fast train from Swansea to London, and for Londoners an alternative way to get from Paddington to southwest England without the drive.

Next Steps

Further engagement with councils, ports, industry partners and potential investors is expected to take place as discussions continue about how the proposed service could progress. We're keen to hear from organisations with expertise in maritime engineering, hydrogen technology, ports, investment and operations who want to help develop the project further.

1. Research Findings

1.1 Survey Response Summary

The online consultation received 4,396 responses since March 2025 and can be found at https://oceanprime.co/projects/ferry/introduction. Respondents were asked whether they would support a zero-emission ferry service between Swansea and South West England, with the option to provide a short comment explaining their position.

The public learned of the research through a single press release, reported in the local press; every member of the public had the same opportunity to respond.

Response	Count	Percentage
Yes – Support the service	4,296	97.7%
Maybe – Conditional support	66	1.5%
No – Do not support	34	0.8%
Total	**4,396**	**100%**

1.2 What Supporters Told Us

Analysis of comments from supportive respondents revealed consistent themes:

Time Savings and Convenience

The most frequently cited benefit was reduced journey time. Respondents noted that driving from Swansea to North Devon or Cornwall via the M4, Prince of Wales Bridge, M5 and A30 routinely takes three to four hours, often longer during holiday periods. A one-hour crossing would transform what is currently an all-day expedition into a practical day trip or short break. As one respondent noted: “Much needed! The drive down to Cornwall/Devon is too long!”

Family and Social Connections

Many respondents have family members living on the opposite side of the channel. The current road journey makes regular visits impractical, particularly for those without cars or who find long drives difficult. A frequent comment pattern was: “I have family in both Swansea and Cornwall. This would be a “gamechanger” for me.”

Environmental Benefits

Strong support emerged for zero-emission transport. Respondents recognised that replacing car journeys with clean maritime transport would reduce pollution and congestion. Comments highlighted that this aligns with wider climate goals: “Great idea to get people off the roads and less emissions.”

Economic Opportunity

Business owners and tourism professionals saw potential for increased visitor numbers and economic activity on both sides of the channel. The service was seen as complementing regeneration efforts in Swansea and tourism development across the region.

Congestion Avoidance

The Brynglas Tunnels on the M4 and the M5 through Bristol were repeatedly cited as sources of frustration. Respondents welcomed an alternative that would bypass these bottlenecks entirely.

1.3 Concerns and Questions Raised

Respondents who voted “No” or “Maybe” raised legitimate questions that will need to be addressed as the project develops:

Commercial Viability

Several respondents cited previous ferry services that had failed, including the Cork ferry and Severn Link. They questioned whether sufficient demand exists to sustain a commercially viable operation without ongoing public subsidy.

Weather and Tidal Conditions

The Bristol Channel has one of the highest tidal ranges in the world [1] and can experience sea conditions similar to mid-Atlantic. In December 2024 Storm Darragh produced waves over 9m. Respondents asked how the service would maintain reliability in adverse weather and whether operations would be seasonal.

Vehicle Transport

While many respondents indicated they would use a foot passenger service, a significant number asked whether the service would carry vehicles; motorbikes, cars and motorhomes. Comments suggested that for some journeys, particularly holidays with luggage or travel with mobility equipment, vehicle transport would be essential.

Destination Infrastructure

Concerns were raised about road access and parking at potential English ports. Respondents from North Devon noted that existing infrastructure in towns such as Ilfracombe was not designed for high volumes of additional traffic.

Price Sensitivity

Respondents wanted assurance that fares would be genuinely affordable and competitive with driving. The target price point of “from £15” has been received positively where mentioned, though some express scepticism that such pricing would prove sustainable.

1.4 Specific Passenger Requirements

Respondents identified several specific requirements that should inform service design:

Foot Passengers: The majority of supportive responses came from those who would travel as foot passengers, particularly for day trips and short breaks where vehicle transport is not required.

Pets: Multiple respondents asked whether dogs and other pets would be permitted on board. This is particularly relevant for tourism-related travel.

Bicycles: Support for carrying bicycles was evident, with respondents seeing potential for cycling holidays and sustainable end-to-end journeys.

Cars and Motorhomes: While foot passenger demand appears strong, a proportion of respondents indicated they would only use the service if vehicles could be transported. This is most relevant for longer holidays and trips requiring luggage.

2. Market Size and Opportunity

2.1 Catchment Population

We used the concept of "close enough for a day trip: The combined population within one hour’s travel of ports across the corridor from Fishguard to Gloucester to Penzance, including the Isles of Scilly, has a resident population of approximately 6 million (ONS mid-2023). These communities also host approximately 20 million overnight tourist visits per year, representing approximately 60 million visitor-days in the corridor. The combined pool of approximately 2.3 billion annual people-day opportunities represents the theoretical total addressable market from which a coastal ferry network would draw passengers.

2.2 Current Cross-Channel Traffic

The two Severn crossings (M48 Severn Bridge and M4 Prince of Wales Bridge) currently carry approximately 80,000 vehicles per day, equivalent to around 29 million vehicle crossings per year [3], representing roughly 40 million passenger journeys based on average vehicle occupancy. Traffic across both crossings has grown by 34% since tolls were removed in

December 2018 [4]. The M48 Severn Bridge has been subject to a 7.5 tonne weight restriction since May 2025, following confirmation of main cable corrosion, with all HGVs diverted to the M4 Prince of Wales Bridge. A signal-controlled system to manage HGV crossings has been delayed by National Highways and is now expected in January 2027, adding further structural pressure to the road network [5]. Bridge traffic is expected to continue to increase.

2.3 The Day Trip Opportunity

A key insight from the research is the potential for day trip travel. Currently, the road journey is too long for practical day visits between Swansea and North Devon or Cornwall. A one-hour crossing would bring destinations within “close enough for a day trip” range, unlocking demand that currently does not exist. This is how Low Cost airlines, like Ryanair and EasyJet grew the aviation market in the last 30 years: they did not merely take passengers from traditional carriers, they created entirely new journeys that were previously impractical or unaffordable.

3. Indicative Pricing and Schedule

3.1 Target Pricing

Based on benchmarking against the economics of low-cost carriers, we are setting a base ticket target price of £15. Tickets will be able to be pre-booked as well as purchased on board for those who fancy a day out or need to travel in an emergency. Car owners will be able to book a travel window, but would be offered the next available sailing where there is space, similar to how Le Shuttle operates.

3.2 Schedule Development

Detailed schedule development will depend on further analysis of tidal access windows at each port, the scope of what port infrastructure upgrades and arrangements are possible, and the expected demand patterns identified through continued engagement. Rugby, football and concerts at Swansea Arena, New Year’s Eve celebrations would all be accessible for people across the channel and those from further along South Wales who don’t wish to drive. The provision of late evening services after concerts have ended are likely to have good support. The target operating model assumes 20 trips per day, operating 350 days per year, (reduced service during periods of extreme weather and waves above 2.5m). The 2.5m figure is the planning assumption for the schedule; the vessel design target for comfortable operations is 3m or more (see 3.3).

Schedule optimisation will prioritise morning and evening services to enable day trips, with pricing designed to spread demand across all services and maximise load factors.

3.3 Concept Ship Design

The design is at an early stage. Development of the vessel design includes building and testing a small-scale prototype. Some key design criteria are as follows:

- 350 Passenger capacity
- 24-32 Car capacity
- Target maximum speed 60 knots, cruising speed 40 knots
- Maximum Draught: 2m
- Target maximum wave height for comfortable operations: 3+m (vessel design target; the schedule assumes reduced service above 2.5m)

- Zero emissions, including CO2, NOX, SOX, particulates, greywater, blackwater, chemical leaching and wash-off. Noise levels cannot be set at this time.
- Hydrogen-Electric power train.

4. Financial Appraisal

Disclaimer: It is important to note these figures and forecasts are based on best estimates and assumptions on what we know today. In many cases targets are based on the understanding that significant research, development and optimal implementation would be required.

4.1 Revenue Potential

The following estimates are illustrative and dependent on vessel design, port infrastructure, fuel availability, pricing and operational assumptions being validated through detailed planning.

Metric	Single Vessel	Full Fleet (8 vessels)
Passenger capacity / year 5 @ 20 one-way trips / day	2,450,000	19,600,000
65% load factor	1,592,500	12,740,000
Ticket revenue (From £15)	£35.0m	£280.3m
Ancillary revenue (£8 avg. / passenger / trip)	£12.7m	£101.9m
Total passenger revenue	**£47.7m**	**£382.2m**

Calculations assume 350 passengers per vessel, 20 one-way services per day, 350 operating days per year. Fleet expansion would occur over five years as routes are established and demand increases.

4.2 Operating Costs

Detailed operating cost analysis is commercially sensitive and will be developed through the business planning process. However, the operating model is designed to achieve significantly lower costs than traditional ferry operations. Exceptionally high-efficiency vessel design, rapid turnarounds with services planned to leave every 20 minutes (mature model) and advanced AI will be used to optimise operations and reduce variable costs. The target is for the service to achieve operating profitability at load factors substantially below those required by traditional ferry operators.

5. Economic Benefits for Swansea and the Region

5.1 Direct Employment

The project would create substantial direct employment in two main areas. First, the manufacturing facility in the Swansea Bay region would employ approximately 350 skilled workers in vessel design, construction and testing. Second, operations require approximately 100 crew members and shore-based staff initially, growing as the fleet and the number of destinations expands. The intention is to recruit and train local staff through agreements with local colleges.

5.2 Supply Chain Employment

Beyond direct employment, the project could generate approximately 1,500 jobs across the supply chain. This includes suppliers of marine systems, materials and components, port services including maintenance and fuel handling, connected transport services such as coaches, taxis and cycle hire, and hospitality and retail serving increased visitor numbers. The intention is to support local food and drink suppliers and develop with them a unique and high quality onboard refreshments offer from each participating port.

5.3 Visitor Economy

The model assumption is 50% of all passengers would be visitors to Swansea and 50% returning residents. At a projected 65% load factor and a single vessel, this would represent up to 800,000 visitors to Swansea annually. Research indicates visitor spend on average is typically £100 per person during their visit, covering food, activities, retail and accommodation. This would generate an estimated £80 million in additional economic activity for the city.

The service as proposed at the time of the research was estimated to bring up to 800,000 visitors a year from a single vessel, and up to £600 million a year in visitor spend from the full fleet of eight vessels, subject to port access limitations, with similar pro rata benefit levels at participating ports across the network. Actual visitor numbers and spend will depend on details of the service yet to be agreed. The benefit to Swansea will depend on the mix of vehicle and foot passengers, the exact disembarkation point, and the local transport services and infrastructure, all of which have yet to be agreed.

5.4 Training and Skills

The project would create demand for new skills in green maritime technology, hydrogen systems and AI-enabled operations. Ocean Prime is committed to developing partnerships with regional colleges to create apprenticeship pathways and training programmes that benefit local residents.

6. Congestion and Pollution Benefits

6.1 Road Traffic Reduction

Passenger journeys transferred from cars to the ferry can reduce traffic on congested road routes. The M4 corridor through South Wales, the Prince of Wales Bridge, and the M5 through Bristol all experience significant congestion, particularly during holiday periods. Transferring even a small proportion of this traffic to maritime transport would reduce pressure on road infrastructure and improve journey reliability for those who continue to drive.

Based on Department for Transport data, the average UK car emits 129 grams of CO_2 per kilometre (207 grams per mile) [6]. A typical journey between Swansea and North Devon by road is approximately 175 miles, producing around 36kg of CO_2 per vehicle.

At projected passenger volumes with a single vessel (1,592,500 passengers/yr) and an average of two passengers per car, this equates to 796,250 car journeys. At 175 miles and 207 g CO_2 per mile (36.2 kg per journey), the maximum saving is approximately 28,900 tonnes of CO_2 per year, assuming every passenger would otherwise have driven. The development of local zero-emission transport connections at each port would encourage more passengers to leave their vehicles behind entirely, potentially increasing these savings further.

6.2 Zero Emission Transport

The proposed vessel concepts are operationally zero-emission from day one, powered by an electric drivetrain with green hydrogen range extension generators. Unlike diesel ferries, there will be no emissions of CO2, SOx, NOx or particulate matter during operation. A zero-emission vessel operating the same route would save approximately 1,800 tonnes of CO2 per year compared to a conventional diesel ferry of equivalent capacity, based on DEFRA greenhouse gas conversion factors for foot passenger ferry travel [7].

Ocean Prime is intending to power all operations using locally generated renewable electricity to minimise the impact on the national grid.

6.3 Air Quality

Maritime transport removes vehicle emissions from densely populated areas. Research from MIT estimates that road transport exhaust is responsible for approximately 3,300 premature deaths annually in the UK [8]. The UK Government's Committee on the Medical Effects of Air Pollutants (COMEAP) estimates that between 28,000 and 36,000 deaths per year in the UK are attributable to long-term exposure to air pollution overall [9]. The economic cost of premature deaths from combustion-related air pollution has been estimated at a minimum of £6 billion per year [8]. By replacing car journeys with zero-emission maritime transport, the service would contribute to improved air quality along the M4/M5 corridor and in the urban areas these routes pass through, including Swansea, Bristol, Cardiff, and Newport. This is particularly significant during peak holiday periods when traffic volumes and associated pollution levels are at their highest.

7. Green Hydrogen: A Strategic Opportunity

The vessels will require green hydrogen. A single vessel operating the planned service frequency would consume approximately 1,000 tonnes of hydrogen per year. With eight vessels at full operation, annual consumption would approach 8,000 tonnes/yr.

This represents a significant opportunity for the emerging green hydrogen sector in Wales and Swansea Bay in particular. Even with a single vessel, OceanJet Line would become one of the largest offtakers of green hydrogen in the UK. This provides a guaranteed demand anchor that could help justify investment in local hydrogen production facilities, creating further jobs and positioning the region as a leader in the green hydrogen economy.

The scale of this demand makes OceanJet Line a strategically important customer for hydrogen producers. We will actively engage with potential suppliers to understand how our requirements could support the development of production capacity in the Swansea Bay area.

8. Four Development Workstreams

Converting the demonstrated public support into an operational service requires detailed work across four interconnected areas:

8.1 Vessel Design

The vessels must be specifically designed for Bristol Channel conditions, achieving speeds of over 40 knots while maintaining passenger comfort in challenging sea states - we do not want anyone to become sea sick. The shallow draught requirement (below 3 metres) to access tidal ports drives a specialist hull design. Further engineering work is required to finalise specifications, conduct tank testing and prepare for construction.

8.2 Port Infrastructure and Local Engagement

Each participating port will require investment in passenger facilities and quay side renewable electricity, fuel storage and management. OceanJet Line is seeking to engage with local councils and port authorities across the entire region. Any port or harbour where there is the potential for customer demand and access arrangements can be provided, will be considered for inclusion in the network, whether for year-round or seasonal services. We encourage councils who wish to explore opportunities for their area to contact Ocean Prime or Swansea Council directly, as route and schedule decisions will be informed by the quality of local partnership arrangements.

8.3 Green Hydrogen and Renewable Electricity

The supply of green hydrogen at predictable prices is fundamental to the business model. We are developing plans for acquiring renewable electricity and green hydrogen to be available at each port we service. This requires engagement with existing local suppliers, planning authorities, technology partners and universities.

8.4 Workforce Development

Operating electric-hydrogen hybrid powered vessels with AI-enabled systems requires new skills that do not widely exist in the UK maritime sector. We are developing ideas for training programmes in partnership with regional colleges and local maritime training providers. This includes the vessel “pilots” who will need to operate one of the most advanced maritime vessels. There is a requirement for customer service and logistics, crew and port management training, plus engineering and technical roles for vessel maintenance, battery and hydrogen systems.

9. Provisional Timeline

The following timeline is a guide for information only: We are aiming for the first vessel to be in service by 2030.

10. Research Conclusion

The research demonstrates overwhelming public support for a zero-emission ferry service connecting Swansea with destinations across the Bristol Channel. With 98% of nearly 4,400 respondents supporting the proposal, the mandate for proceeding to detailed planning is clear.

The economic case is compelling. The service would create hundreds of direct jobs and support thousands more across the supply chain, generate substantial visitor economy benefits for Swansea and all participating ports, reduce congestion and pollution on overcrowded road routes, and position the region at the forefront of green maritime technology.

The challenges are real but manageable. Weather and tidal conditions require specialist vessel design. Port infrastructure requires investment. Hydrogen supply chains need development. However, these are engineering and commercial challenges rather than fundamental barriers.

We are grateful for the partnership with Swansea Council that has enabled this research, and we look forward to continued collaboration as we work to make this vision a reality.

For further information:

Dave Sampson, CEO - Ocean Prime Industries Ltd

References

[1] ABP South Wales, “Bristol Channel Tides.” The Bristol Channel / Severn Estuary has a mean spring tidal range of 12.3m at Avonmouth. The Bay of Fundy, Canada (16.3m) is the largest; some sources place Ungava Bay, Canada as comparable, making the Bristol Channel the second or third largest globally. See also: Gao, C. & Adcock, T.A.A., “On the Tidal Resonance of the Bristol Channel,” International Journal of Offshore and Polar **Engineering, 2017.**

[2] ONS Mid-2023 Population Estimates for England and Wales (revised July 2024). The resident population covers all local authorities within the corridor from Pembrokeshire to Gloucestershire on the Welsh/English side, and from Gloucestershire to Cornwall including the Isles of Scilly on the English side. Tourism data from VisitBritain, England Domestic Overnight Trips and Day Visits: Subnational Data (county-level two-year averages, October 2021 to September 2023); Visit Cornwall, Tourism Trends 2024 (19.3 million day and staying trips in 2019); Visit West, Regional Visitor Economy Research 2022; Welsh Government, Domestic GB Tourism Statistics 2023 (overnight trips and day visits to Wales). Visitor-days calculated as overnight trips multiplied by average length of stay (Great Britain Tourism Survey). People-day opportunities represent the sum of resident-days (population × 365), overnight visitor-days, and non-resident day visits, and are a theoretical measure of the total addressable market, not a forecast of demand.

[3] National Highways, April 2025. The M48 Severn Bridge carried approximately 32,000 vehicles per day in 2024. The Institution of Civil Engineers states the two crossings are used by more than 80,000 vehicles per day. GOV.UK press release (July 2017) confirmed the bridges were “used by more than 25 million vehicles each year” prior to toll removal; traffic has since increased by 34%.

[4] National Highways, April 2025: “Since the tolls were removed on the bridge in 2018 there has been a 34% increase in traffic.”

[5] National Highways, May 2025. Weight restrictions (7.5 tonnes maximum) introduced on M48 Severn Bridge following main cable corrosion confirmed by assessment finalised December 2024. HGVs diverted to M4 Prince of Wales Bridge. National Highways confirmed in September 2025 that a signal-controlled system to manage HGV crossings was expected by October 2026. National Highways has since delayed the system to January 2027 to allow additional testing.

[6] Department for Transport, Vehicle Emissions data 2024: average UK car CO2 emissions are 129.4 grams per km (207.1 grams per mile). Based on DEFRA/DESNZ Greenhouse Gas Reporting: Conversion Factors 2024.

[7] DEFRA/DESNZ Greenhouse Gas Reporting: Conversion Factors 2024. Foot passenger ferry emission factor: 0.01874 kg CO2e per passenger-km. Applied to approximately 60km crossing distance and 1,592,500 annual passengers at 65% load factor gives approximately 1,800 tonnes CO2 for an equivalent diesel-powered service.

[8] Yim, S.H.L. and Barrett, S.R.H., "Public Health Impacts of Combustion Emissions in the United Kingdom," Environmental Science & Technology, 2012, 46(8): 4291-4296. Road transport exhaust was the single greatest contributor to premature death from combustion emissions, affecting approximately 3,300 people per year. The economic cost of all UK combustion-related premature deaths was estimated at "at least £6 billion a year."

[9] Committee on the Medical Effects of Air Pollutants (COMEAP), "Associations of long-term average concentrations of nitrogen dioxide with mortality," 2018 (updated 2022). COMEAP estimates between 28,000 and 36,000 premature deaths per year in the UK are attributable to long-term exposure to air pollution (PM2.5 and NO2 combined).

Important information and limitations

This report records the views of those who responded to the public consultation. Participation was voluntary; the results should not be interpreted as a statistically representative sample of the population or as a binding public mandate.

Forecast passenger numbers, fares, revenue, costs, economic benefits, emissions savings, vessel performance and development timescales are estimates or targets based on assumptions at the time of preparation. They are illustrative, not guarantees, and remain subject to research, engineering validation, commercial feasibility, port access, funding and applicable consents.

Routes, ports, services, schedules and prices have not been finalised. Publication does not constitute a commitment by Ocean Prime Industries Ltd, Swansea Council or any other organisation to develop, fund or operate a particular service.

This report is provided for information and discussion. It is not an offer of securities, an invitation to invest or investment advice.

The published report excludes separately held survey datasets, supporting spreadsheets, proprietary internal financial models, detailed cost analyses and commercially sensitive assessments. Such material is not publicly available. Publication of this report does not authorise access to, use of or disclosure of those materials; any disclosure is solely at the discretion of Ocean Prime Industries Ltd, subject to prior written authorisation and appropriate confidentiality safeguards.

APPENDICES

Public report and confidential supporting material. This report is intended for public release, but its supporting appendices and source files are not. These include proprietary internal financial models, operating-cost analyses, commercial assumptions, spreadsheets, survey-response datasets and detailed port assessments. The supporting files are held separately by Ocean Prime Industries Ltd, are confidential and commercially sensitive, and are expressly excluded from this public report. Publication of this report does not authorise the publication, redistribution, access to or disclosure of any supporting files. Ocean Prime Industries Ltd retains sole discretion over any separate disclosure, subject to prior written authorisation and appropriate confidentiality and data-protection safeguards.

Appendix A: Survey Response Data

Full anonymised survey responses are retained as a separate, non-public data file. Comments have been analysed thematically; individual responses are retained for reference but are not published to protect respondent privacy.

Appendix B: Financial Model Assumptions

The financial projections in this report are based on the following key assumptions:

- Vessel capacity: 350 passengers and 32 cars
- Operating days: 350 per year (weather adjusted)
- Services per day: 20 trips
- Base ticket price: £15
- Average ancillary revenue: £8 per passenger
- Target load factor (Year 5): 65%
- Mature fleet size (Year 5): 8 vessels

All projections are dependent on vessel design enabling required draught specifications, port infrastructure enabling tidal access, and weather assumptions proving accurate. The model does not account for vehicle revenue, which would provide additional revenue if vehicle operations can be introduced along the Somerset, Devon or Cornwall coast

Appendix C: Detailed Cost Analysis

The detailed operating-cost breakdown and underlying financial spreadsheets are proprietary, confidential commercial information belonging to Ocean Prime Industries Ltd. They are not included in this public report and are not available for public access. Any limited disclosure to a third party would require Ocean Prime Industries Ltd's prior written authorisation and an appropriate confidentiality agreement. No entitlement to inspect or receive these materials is created by this report.

Appendix D: Port Assessment Criteria

Ports are assessed against the following criteria for inclusion in the network: tidal access window and draught restrictions, existing infrastructure and upgrade requirements, road and rail connectivity, parking availability, local authority support, and demonstrable customer demand from the catchment area. Assessments are ongoing and new ports may be added to the network as arrangements are confirmed.